

Cloud Data Engineering

Lecture 2

Lecture Agenda

- What is big-data?
- History of cloud computing
- Types of clouds
- Data-engineering
- Python libraries for data science and cloud computing

What is Big Data?

Big data is a collection of large datasets that cannot be processed using traditional computing techniques. **It is not a single technique or a tool.**

Benefits

- Using the information kept in the social network like Facebook, twitter, etc.
- Using the information in the social media like preferences and product perception of their consumers.

Sources of Big Data

Social networking sites: Facebook, Google, LinkedIn

E-commerce site: Amazon, Flipkart, Alibaba

Weather Station: All the weather station and satellite gives very huge data which are stored and manipulated to forecast weather

Telecom company: Airtel, Vodafone

Share Market: Stock exchange across the world

3V's of Big Data

Velocity: The data is increasing at a very fast rate.

It is estimated that the volume of data will double in every 2 years.

Variety: Data is structured as well as unstructured.

structured – predefined schema

unstructured – each document has different properties

Volume: The amount of data which we deal with is of very large size of **Peta** bytes.

(1 Peta = 1024 TB)

Handling Big Data

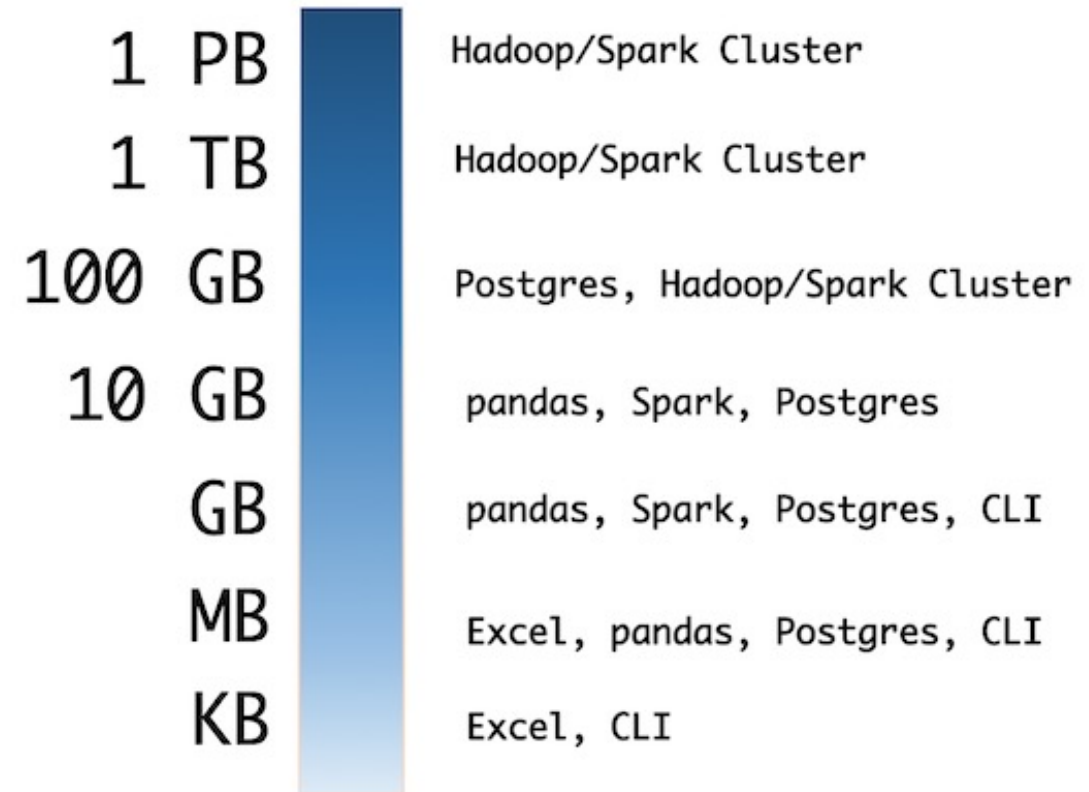
Excel

Pandas – Lecture 3

Hadoop – Lectures 4,5

Spark – Lectures 6,7

Streaming – Lectures 10,11



What is Cloud?

Datacenter hardware and software that the vendors use to offer the computing resources and services.



Cloud Computing

- Represents both the cloud & the provided services
- Why call it “cloud computing”?
 - Some say because the computing happens out there "in the clouds" 😊
 - Wikipedia: "the term derives from the fact that most technology diagrams depict the Internet or IP availability by using a drawing of a cloud."

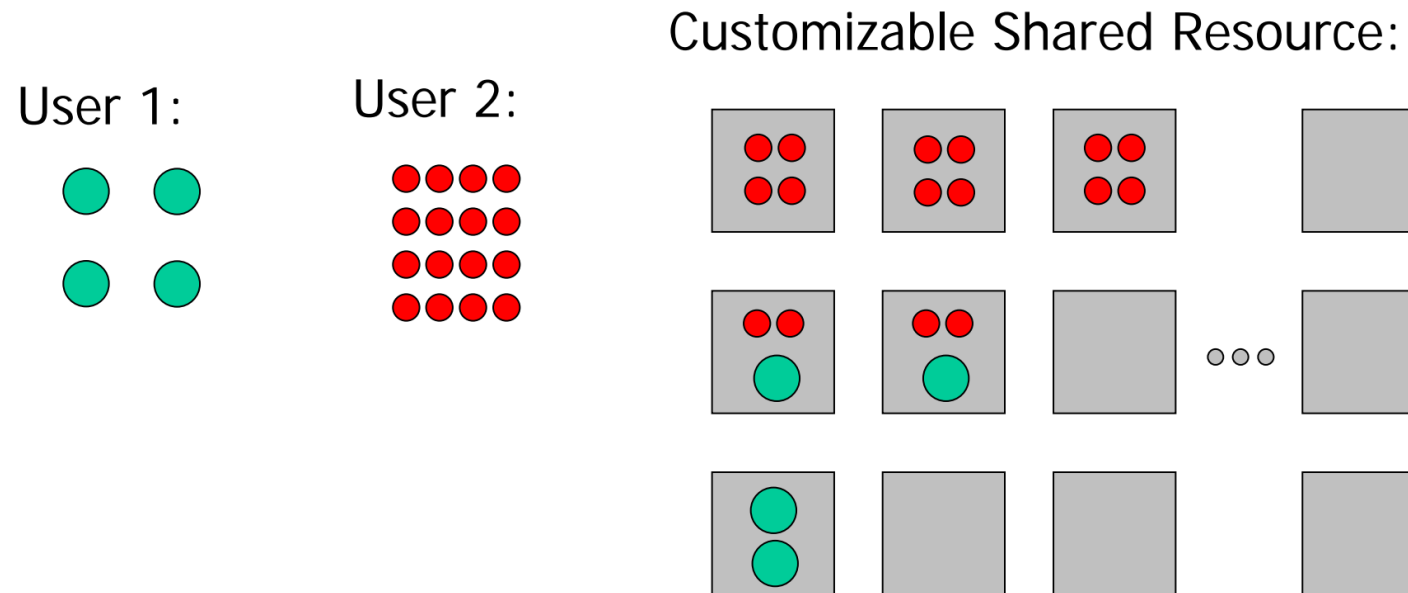
Cloud Computing Services

- **Software as a Service (SaaS)** – applications through the browser
- **Platform as a Service (PaaS)** - Delivery of a computing platform for custom software development as a service
- **Infrastructure as a Service (IaaS)** - Delivery of computer infrastructure as a service
- And more ..



Resource Sharing

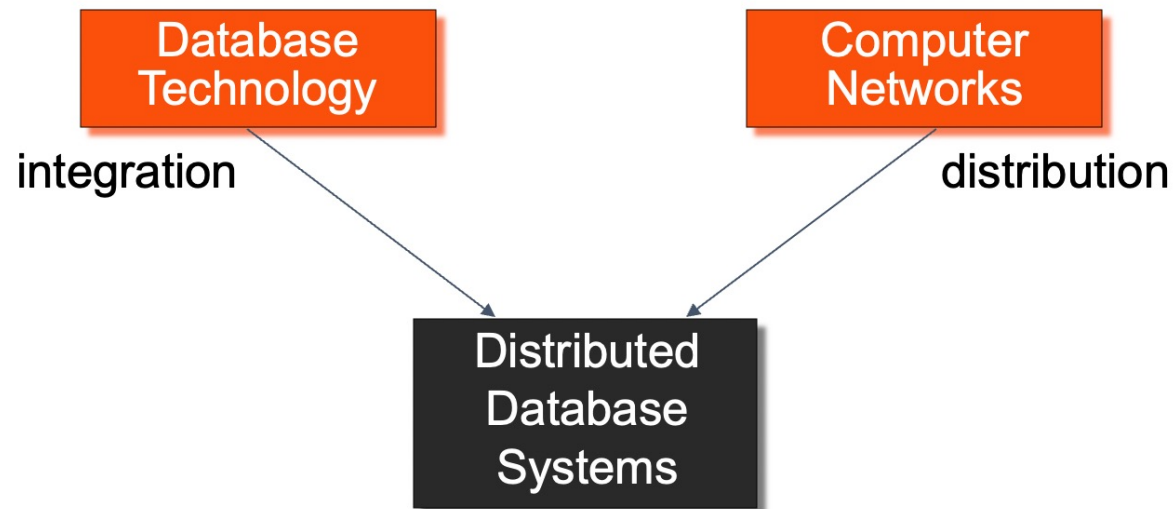
Offering computing resources as a service or utility through **virtualization**



Distributed Computing

Multiple processing elements that are interconnected by a computer network and that cooperate in performing their assigned tasks.

What is being distributed? Processing logic, Function, Data, Control

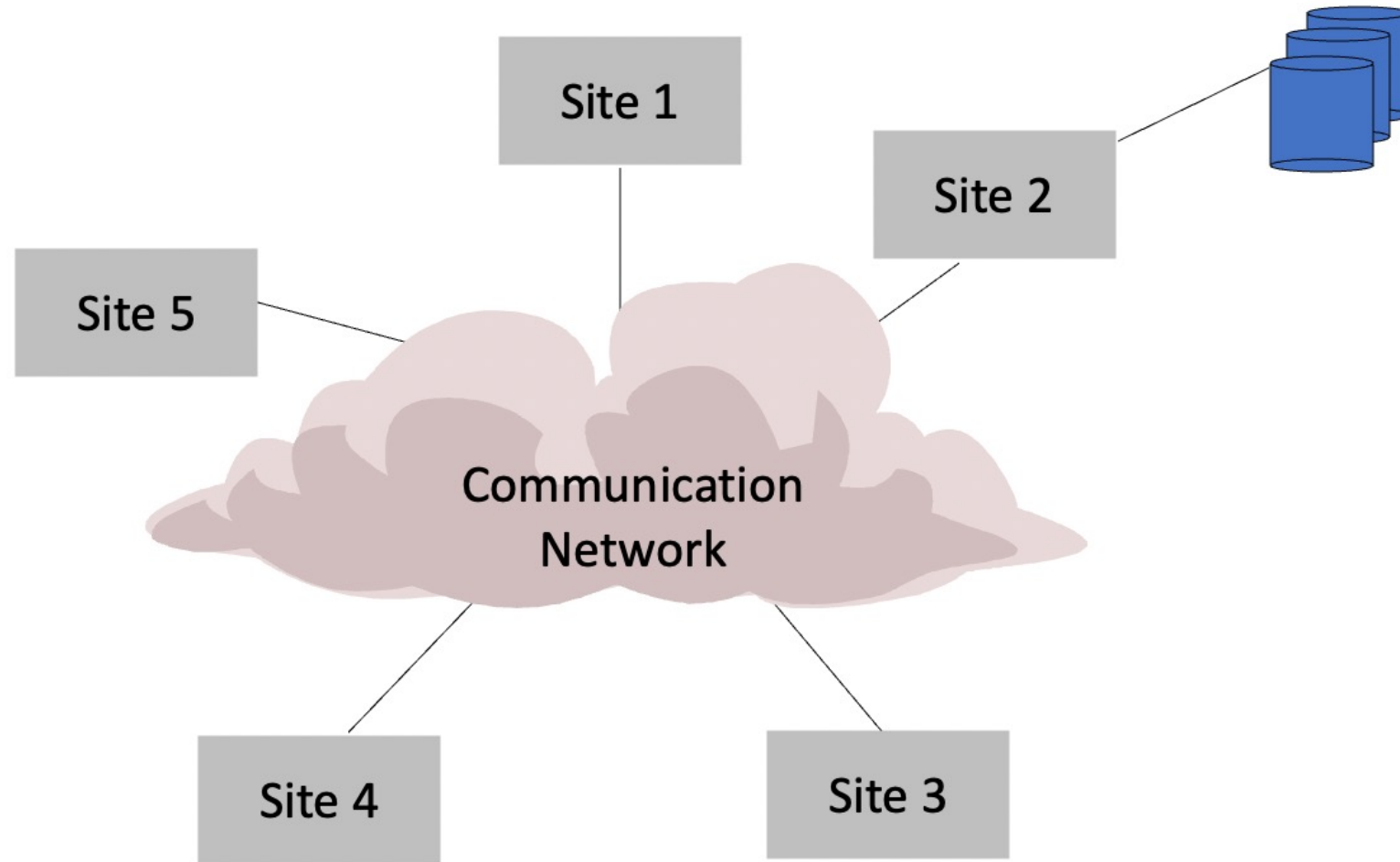


Distributed Database System

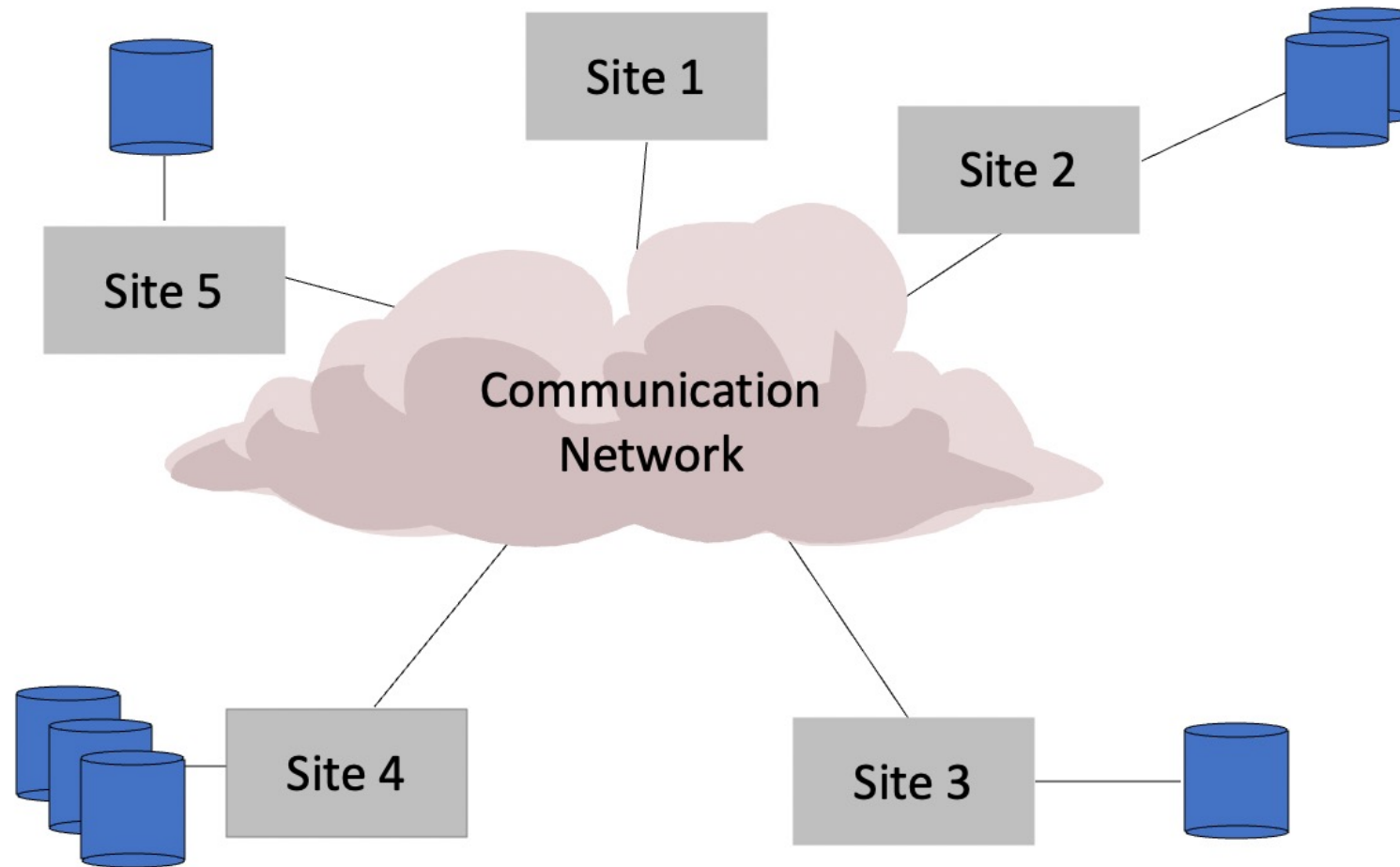
A **distributed database** (DDB) is a collection of multiple, logically interrelated databases distributed over a computer network.

A distributed database management system (D–DBMS) is the software that manages the DDB and provides an access mechanism that makes this distribution transparent to the users.

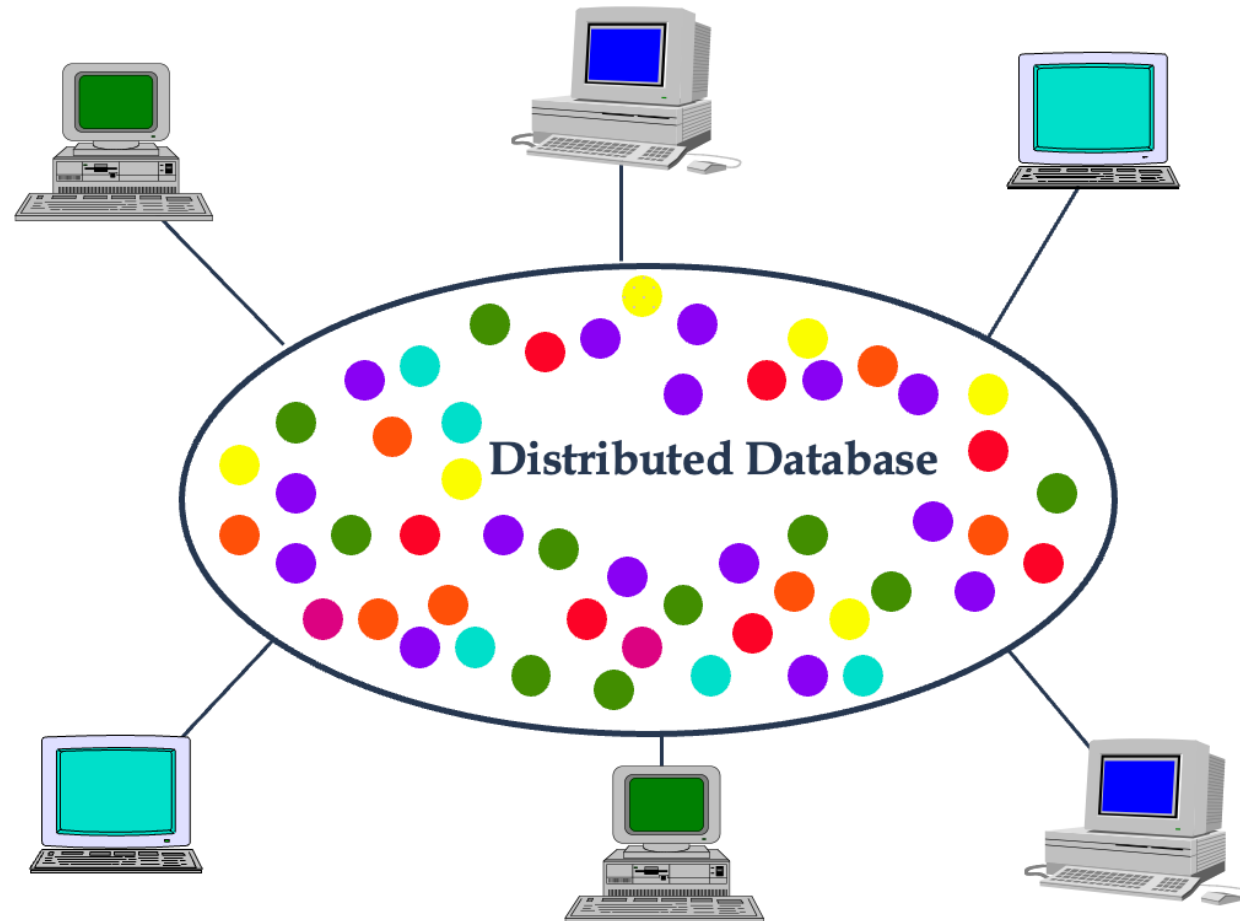
Centralized DBMS



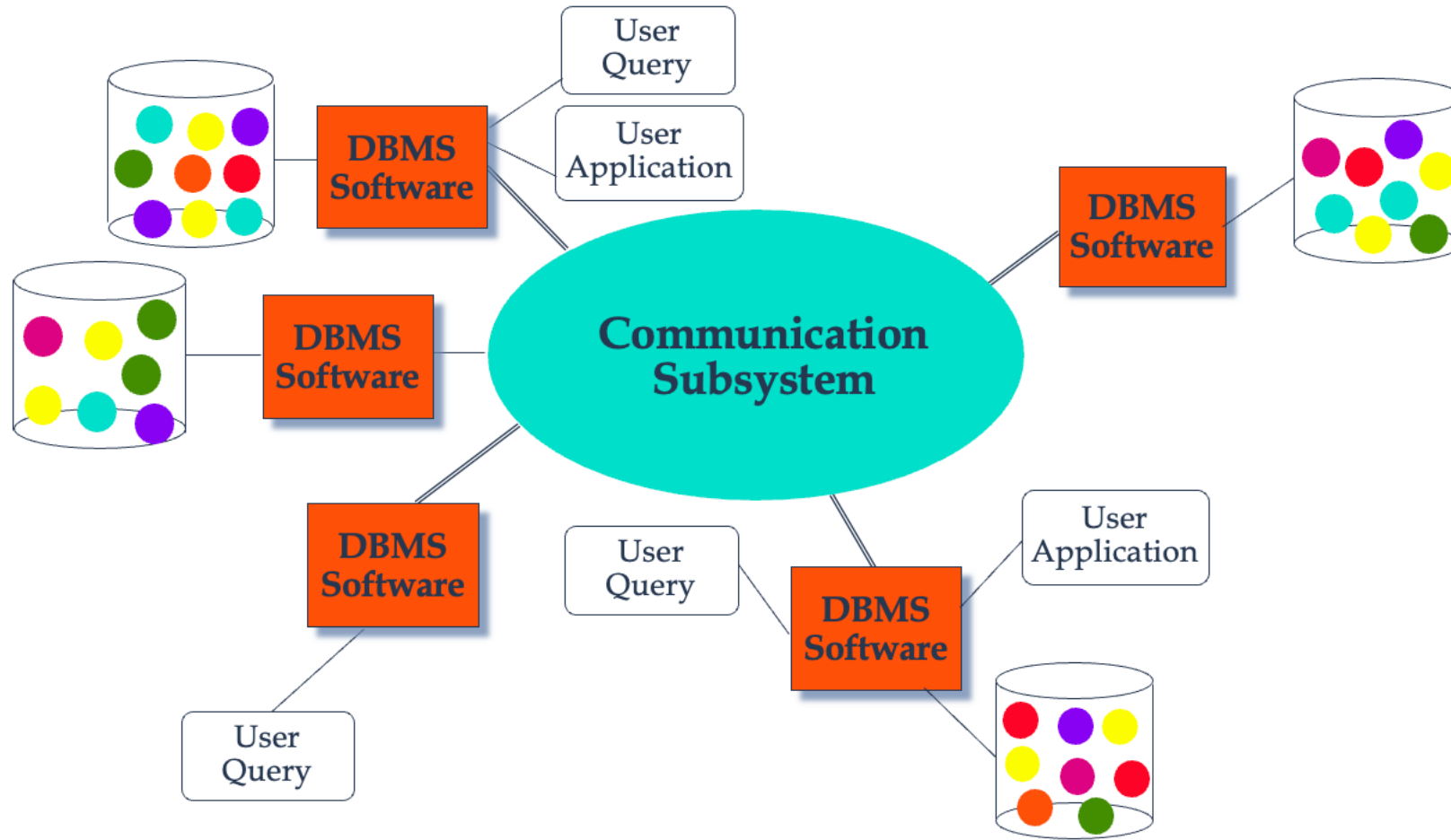
Distributed DBMS



Distributed Database – User View



Distributed Database – Reality



Distributed DBMS Issues

- **Distributed Database Design**

- How to distribute the database
- Replicated & non-replicated database distribution
- A related problem in directory management

- **Query Processing**

- Convert user transactions to data manipulation instructions
- Optimization problem
 - $\min\{\text{cost} = \text{data transmission} + \text{local processing}\}$

Distributed DBMS Issues

- **Concurrency Control**

- Synchronization of concurrent accesses
- Consistency and isolation of transactions' effects
- Deadlock management

- **Reliability**

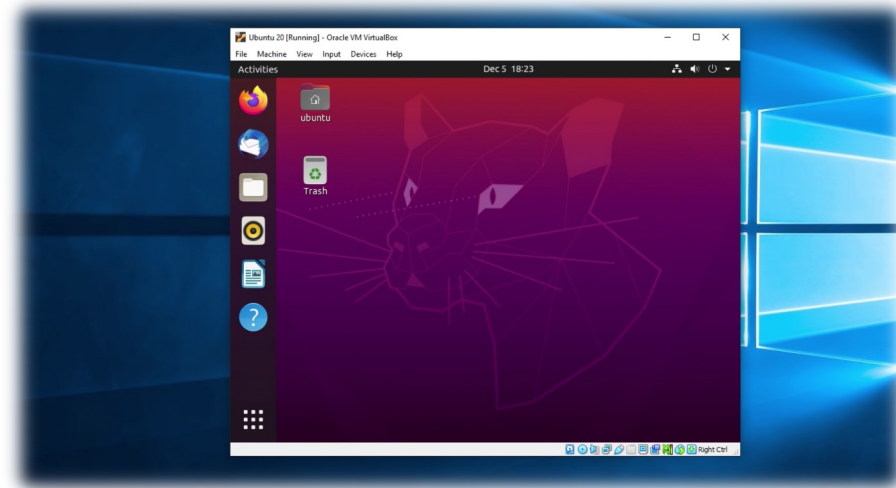
- How to make the system resilient to failures
- Atomicity and durability

Virtual Machines - VMs

- Virtual machines (VMs) offer novel solutions to underutilized resources, application inflexibility, software manage ability, and security concerns in existing physical machines.
- Today, to build large clusters, grids, and clouds, we need to access large amounts of computing, storage, and networking resources in a virtualized manner.
- We need to **aggregate** those resources, and hopefully, offer a single system image.
- A cloud of provisioned resources must rely on virtualization of processors, memory, and I/O facilities dynamically.

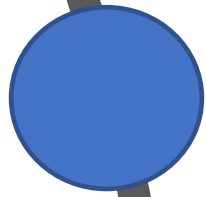
Virtual Machines Operations

- The VMs can be multiplexed between hardware machines
- A VM can be suspended and stored in stable storage
- A suspended VM can be resumed or provisioned to a new hardware platform
- A VM can be migrated from one hardware platform to another

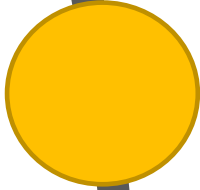


Python Libraries for Data Science

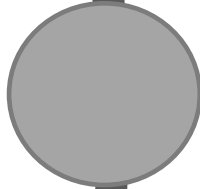
Tutorial Content



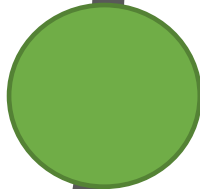
Overview of Python Libraries for Data Scientists



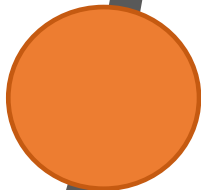
Reading Data; Selecting and Filtering the Data; Data manipulation, sorting, grouping, rearranging



Plotting the data



Descriptive statistics



Inferential statistics

Python Libraries for Data Science

Many popular Python toolboxes/libraries:

- NumPy
- SciPy
- Pandas
- SciKit-Learn

Visualization libraries

- matplotlib
- Seaborn



*All these libraries are
installed on the SCC*

Python Libraries for Data Science

NumPy:

- introduces objects for multidimensional arrays and matrices, as well as functions that allow to easily perform advanced mathematical and statistical operations on those objects
- provides vectorization of mathematical operations on arrays and matrices which significantly improves the performance
- many other python libraries are built on NumPy

Python Libraries for Data Science

SciPy:

- collection of algorithms for linear algebra, differential equations, numerical integration, optimization, statistics and more
- part of SciPy Stack
- built on NumPy

Python Libraries for Data Science

Pandas:

- adds data structures and tools designed to work with table-like data (similar to Series and Data Frames in R)
- provides tools for data manipulation: reshaping, merging, sorting, slicing, aggregation etc.
- allows handling missing data

Python Libraries for Data Science

SciKit-Learn:

- provides machine learning algorithms: classification, regression, clustering, model validation etc.
- built on NumPy, SciPy and matplotlib

Python Libraries for Data Science

matplotlib:

- python 2D plotting library which produces publication quality figures in a variety of hardcopy formats
- a set of functionalities similar to those of MATLAB
- line plots, scatter plots, bar-charts, histograms, pie charts etc.

Python Libraries for Data Science

Seaborn:

- based on matplotlib
- provides high level interface for drawing attractive statistical graphics
- Similar (in style) to the popular ggplot2 library in R

Advantages of Pandas

- Data representation
 - It represents data in a form that is very much suited for data analysis through its Dataframe and Series data structures.
- Data subsetting and filtering
 - It provides for easy subsetting and filtering of data. It provides procedures that are suited for data analysis.
- Concise and clear code
 - It provides functionality to write clear and concise code. It allows us to focus on the task at hand, rather than have to write tedious code.

Numpy & Pandas Tutorial